Practical No :8-C

Aim: Create a BroadcastReceiver that listens for the Android Intent ACTION_BATTERY_CHANGED broadcast and shows a Toast displaying the current battery percentage. code in android

This BroadcastReceiver listens for battery change events and displays the current battery percentage using a Toast message.

Step 1: Create BatteryReceiver.java class file

BatteryReceiver.java

```
package com.example.batteryapp;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.os.BatteryManager;
import android.widget.Toast;

public class BatteryReceiver extends BroadcastReceiver {
   @Override
   public void onReceive(Context context, Intent intent) {
      int level = intent.getIntExtra(BatteryManager.EXTRA_LEVEL, -1);
      int scale = intent.getIntExtra(BatteryManager.EXTRA_SCALE, -1);
      int batteryPercent = (level * 100) / scale;
      Toast.makeText(context,
            "Battery Percentage: " + batteryPercent + "%",
            Toast.LENGTH_LONG).show();
   }
}
```

Step 2: Register Receiver in MainActivity (Dynamic Registration)

MainActivity.java

```
package com.example.batteryapp;
import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
  BatteryReceiver batteryReceiver;

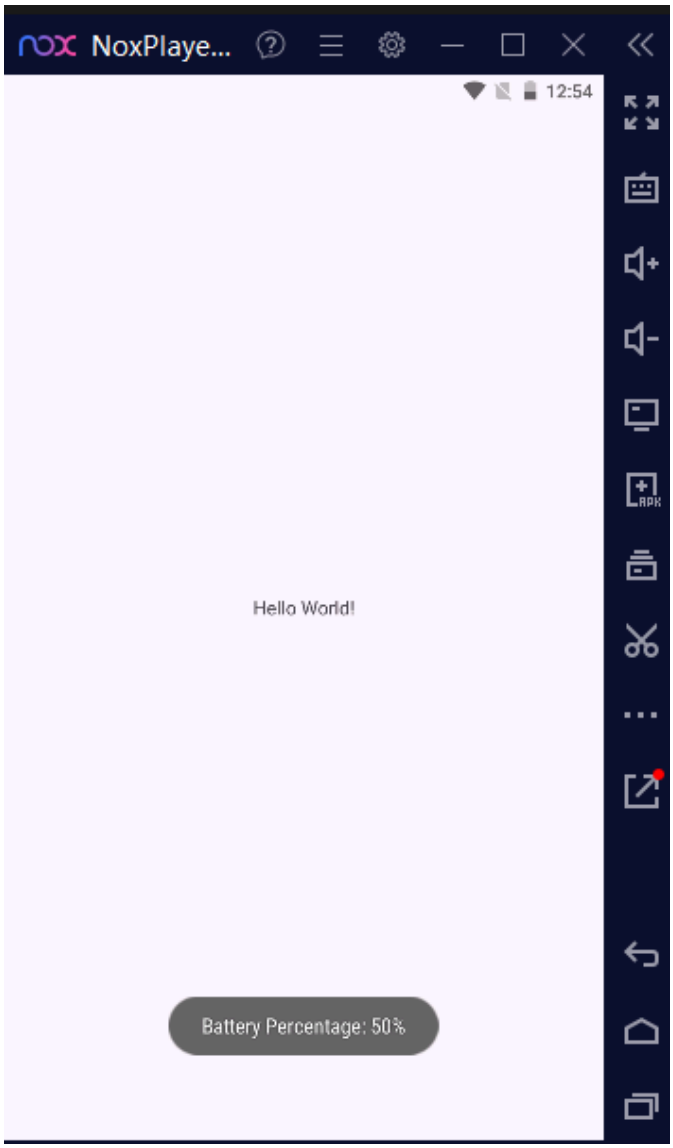
  @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        batteryReceiver = new BatteryReceiver();

        IntentFilter intentFilter = new IntentFilter(Intent.ACTION_BATTERY_CHANGED);
        registerReceiver(batteryReceiver, intentFilter);
    }
}
```

Output:



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